

Branch & Bound(Racine,Sol,Cout)

$\forall i \text{ val}(i) \leftarrow +\infty; \text{Pere}(i) \leftarrow \perp;$

$\text{Cout} \leftarrow +\infty; \text{Sol} \leftarrow \{\};$

$V \leftarrow \{\text{Racine}\}; \text{Val}(\text{Racine}) \leftarrow 0;$

tant que $V \neq \{\}$ **faire**

$n \leftarrow \text{prendreUnElement}(V); V \leftarrow V \setminus \{n\};$

si $(\text{Val}(n) < \text{Cout})$ **alors**

pour chaque (fils f de n) **faire**

si $(\text{Val}(f) > \text{Val}(n) + k(f,n))$ **alors**

$\text{Val}(f) \leftarrow \text{Val}(n) + k(f,n); \text{Pere}(f) \leftarrow n;$

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si (f est un nœud terminal) **alors**

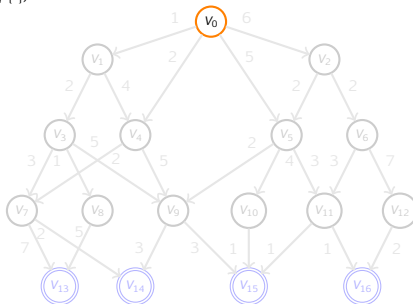
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n (Val(n))	f (Val(f))	Pere(f)	V

$\text{Cout} = -, \text{Sol} = -$



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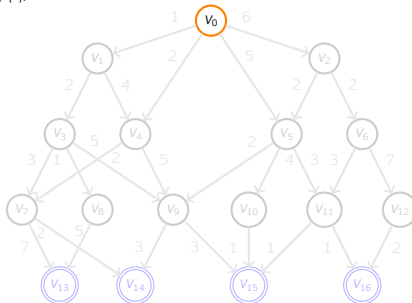
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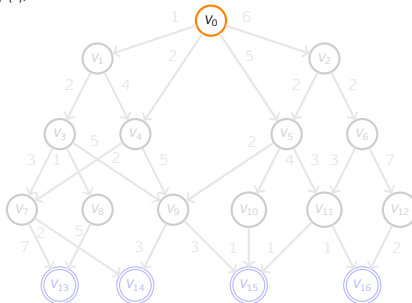
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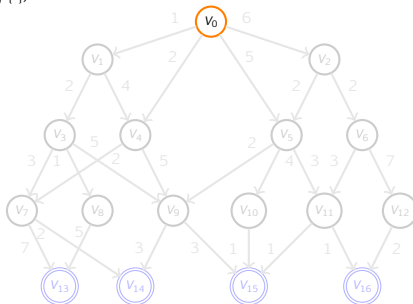
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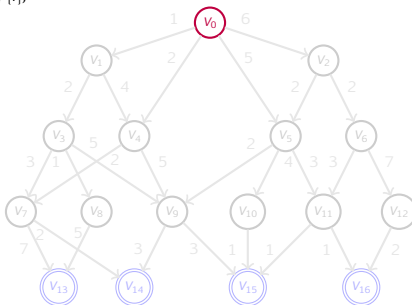
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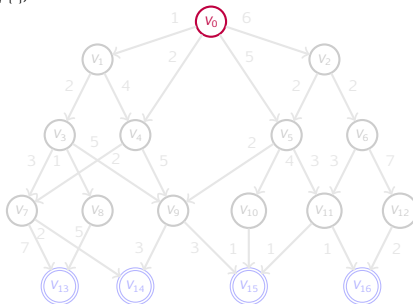
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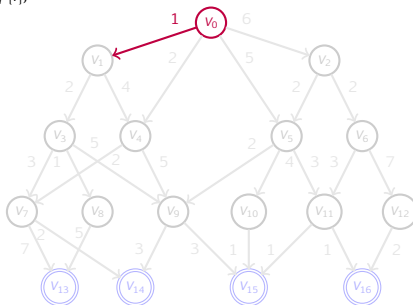
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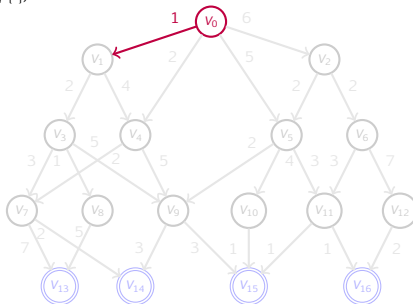
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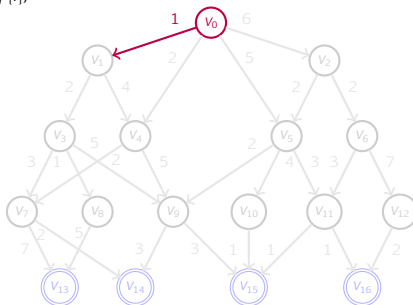
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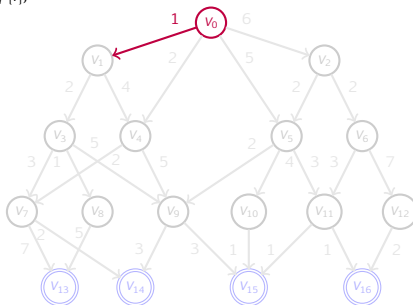
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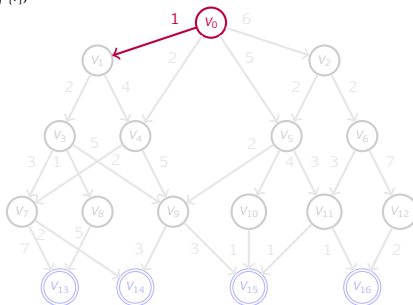
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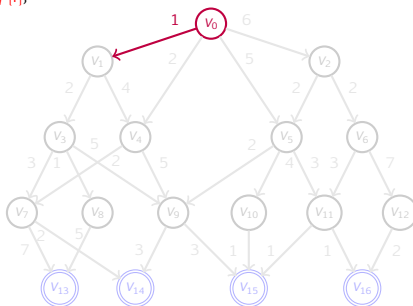
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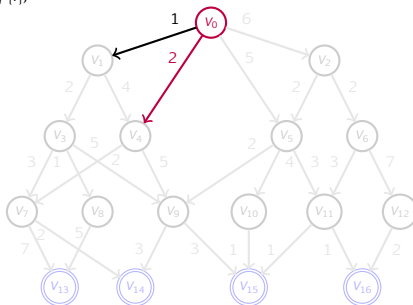
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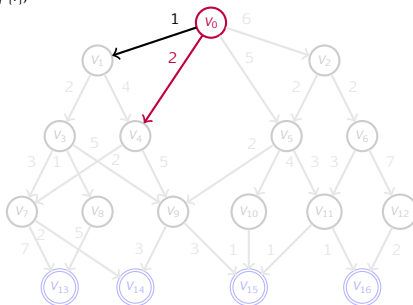
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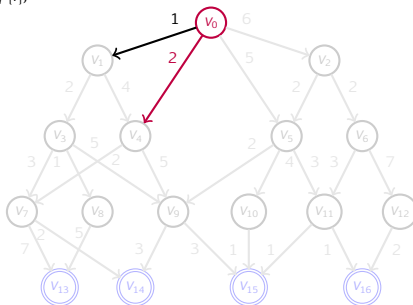
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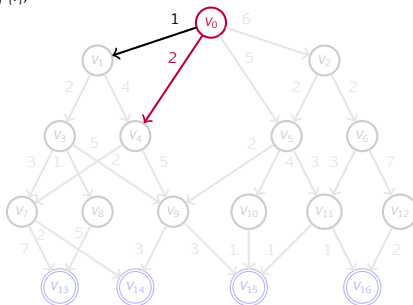
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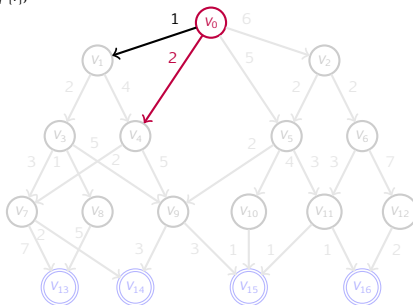
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$\text{Cout} = +\infty, \text{Sol} = \{\}$



Branch & Bound(Racine,Sol,Cout)

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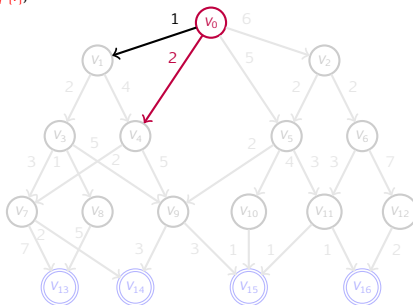
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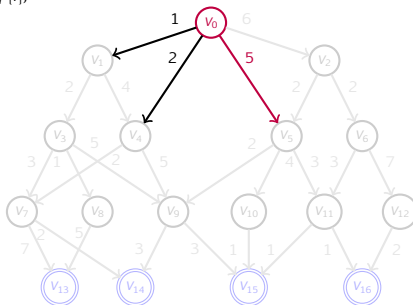
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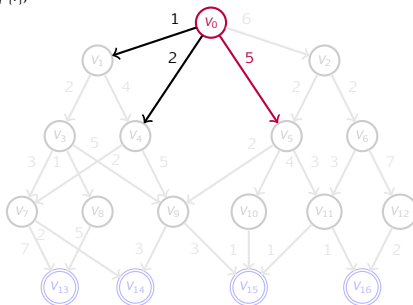
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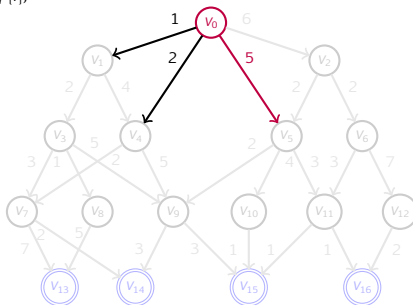
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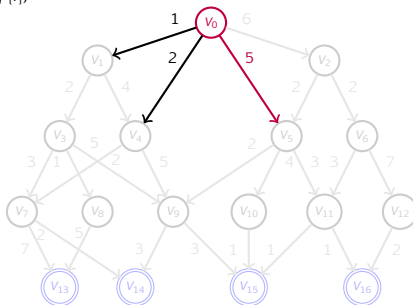
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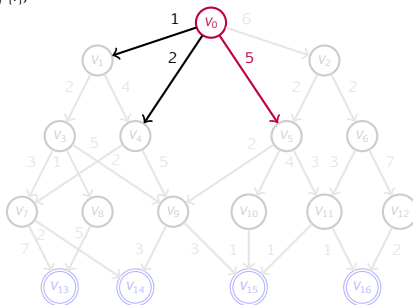
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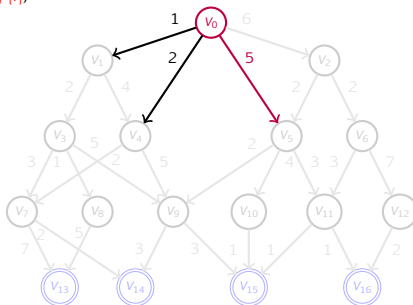
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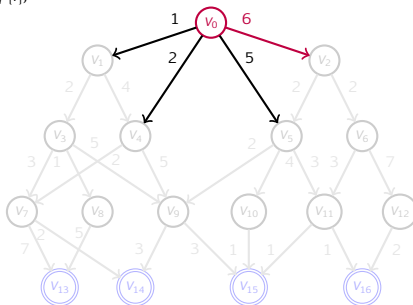
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	V ₅ (5)	V ₀	{V ₁ , V ₄ , V ₅ }
	V ₂ (0)	⊥	{V ₁ , V ₄ , V ₅ }

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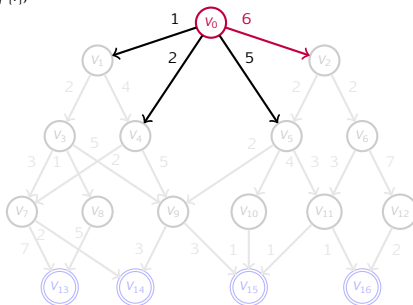
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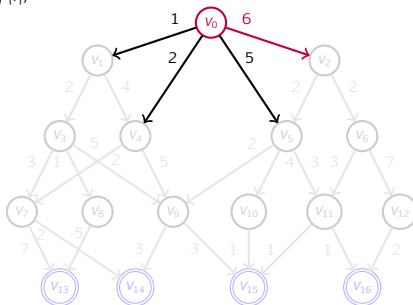
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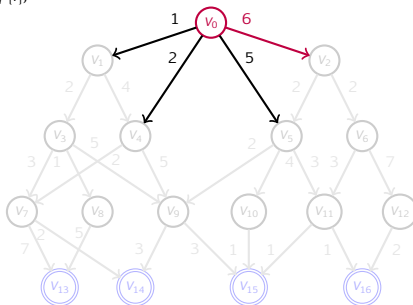
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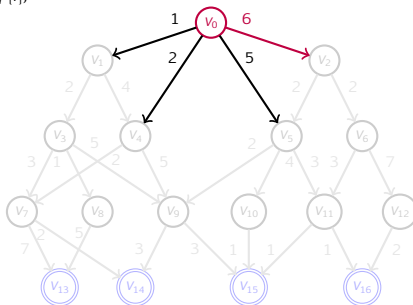
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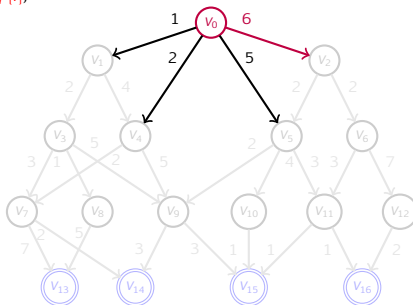
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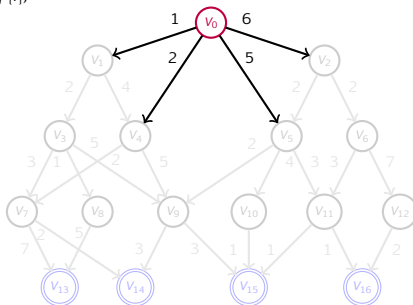
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$\text{Val}(f) \leftarrow \text{Val}(n) + k(f,n); \text{Pere}(f) \leftarrow n;$

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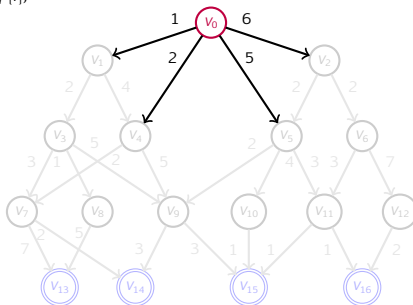
$\text{Cout} \leftarrow \text{Val}(f);$

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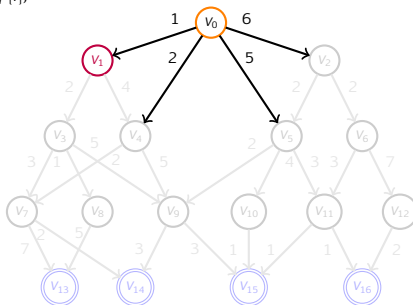
sinon $V \leftarrow V \cup_n \{f\};$

n (Val(n))	f (Val(f))	Pere(f)	V
$V_0(0)$	$V_1(1)$	V_0	$\{V_1\}$
	$V_4(2)$	V_0	$\{V_1, V_4\}$
	$V_5(5)$	V_0	$\{V_1, V_4, V_5\}$
	$V_2(6)$	V_0	$\{V_1, V_4, V_5, V_2\}$

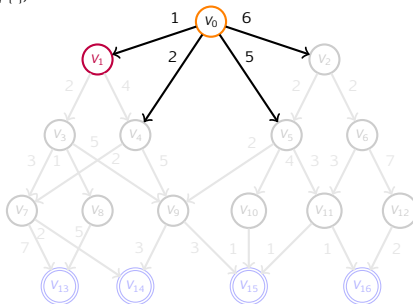
$\text{Cout} = +\infty, \text{Sol} = \{\}$



n (Val(n))	f (Val(f))	Pere(f)	V
V ₁ (1)	()		{V ₄ , V ₅ , V ₂ }

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Branch & Bound(Racine,Sol,Cout)

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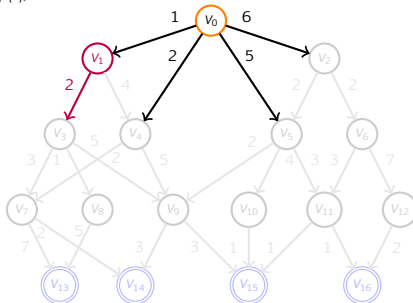
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$V_1(1)$	$V_3(0)$	$\perp$	$\{V_4, V_5, V_2\}$

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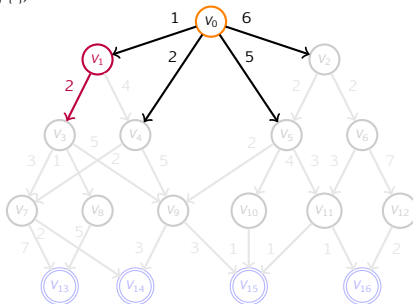
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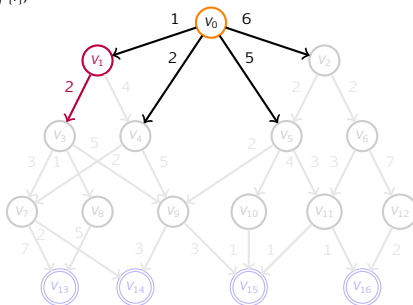
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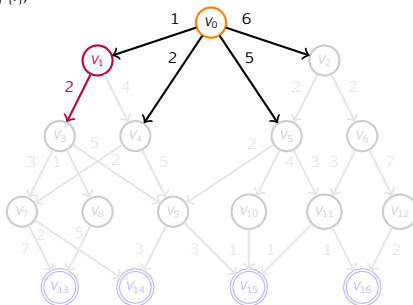
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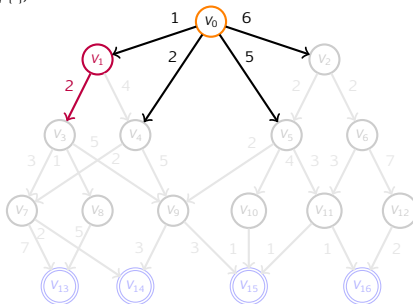
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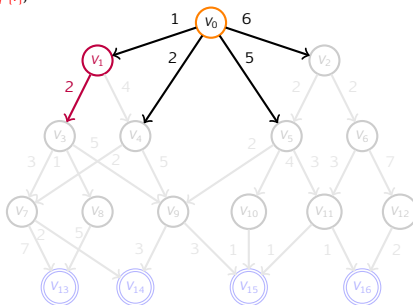
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n (Val(n))	f (Val(f))	Pere(f)	V
V ₁ (1)	V ₃ (3) V ₄ (2)	V ₁ V ₀	{V ₄ , V ₃ , V ₅ , V ₂ } {V ₄ , V ₃ , V ₅ , V ₂ }

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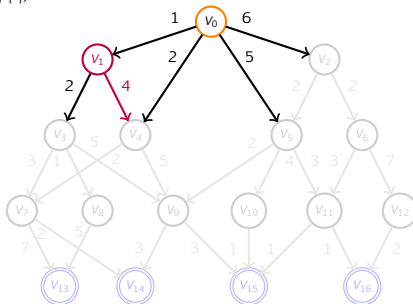
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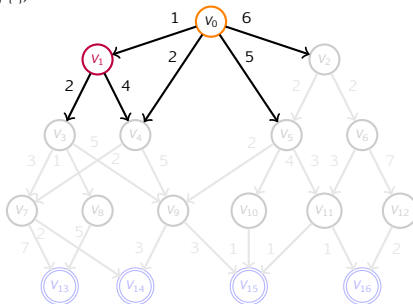
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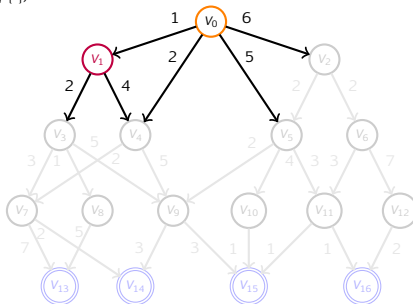
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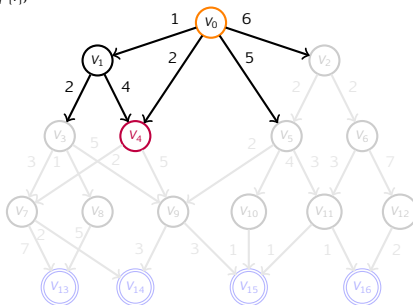
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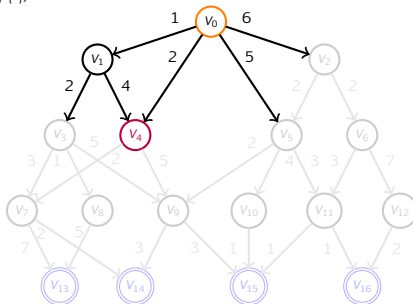
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n (Val(n))	f (Val(f))	Pere(f)	V
V ₄ (2)	V ₇ (4)	⊥	{V ₃ , V ₅ , V ₂ }

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$$n \leftarrow \text{prendreUnElement}(V); V \leftarrow V \setminus \{n\};$$

si ($Val(n) < Cout$) alors

pour chaque (fils f de n) faire

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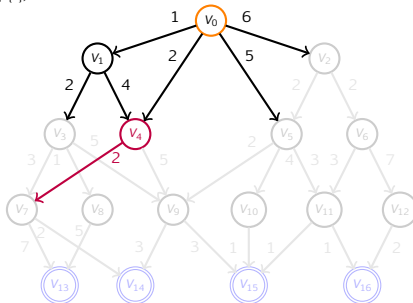
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si (f est un nœud terminal) alors

Cout $\leftarrow$ Val(f);

```
Sol ← cheminMenantA(f);
```

$$\text{sinon } V \leftarrow V \cup_N \{f\};$$


n (Val(n))	f (Val(f))	Pere(f)	V
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si ($Val(n) < Cout$) alors

pour chaque (fils f de n) faire

si $(Val(f) > Val(n) + k(f,n))$ alors

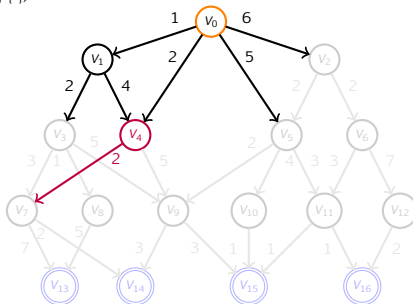
$$\text{Val}(f) \leftarrow \text{Val}(n) + k(f,n); \text{Pere}(f) \leftarrow n;$$

si $(Val(f) < Cout)$ alors

si (f est un nœud terminal) alors

Cout $\leftarrow$ Val(f);

```
Sol ← cheminMenantA(f);
```

$$\text{sinon } V \leftarrow V \cup_n \{f\};$$


n (Val(n))	f (Val(f))	Pere(f)	V
V ₄ (2)	V ₇ (4)	V ₄	{V ₃ , V ₅ , V ₂ }

$$\text{Cout} = +\infty, \text{Sol} = \{\}$$
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n (Val(n))	f (Val(f))	Pere(f)	V
V ₄ (2)	V ₇ (4)	V ₄	{V ₃ , V ₅ , V ₂ }

$$\text{Cout} = +\infty, \text{Sol} = \{\}$$
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si ($Val(n) < Cout$) alors

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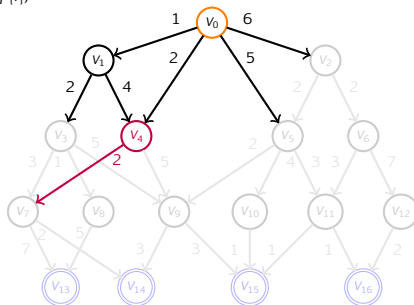
si $(Val(f) < Cout)$ alors

si (*f* est un nœud terminal) alors

Cout $\leftarrow$ Val(f);

Sol ← cheminMenantA(f);

```
sinon  $V \leftarrow V \cup_n \{f\};$ 
```



n (Val(n))	f (Val(f))	Pere(f)	V
V ₄ (2)	V ₇ (4)	V ₄	{V ₃ , V ₇ , V ₅ , V ₂ }

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n (Val(n))	f (Val(f))	Pere(f)	V
V ₄ (2)	V ₇ (4) V ₉ (7)	V ₄ ⊥	{V ₃ , V ₇ , V ₅ , V ₂ } {V ₃ , V ₇ , V ₅ , V ₂ }

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Branch & Bound(Racine,Sol,Cout)

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tant que $V \neq \{\}$ **faire**

$n \leftarrow \text{prendreUnElement}(V); V \leftarrow V \setminus \{n\};$

si $(\text{Val}(n) < \text{Cout})$ **alors**

pour chaque (fils f de n) **faire**

si $(\text{Val}(f) > \text{Val}(n) + k(f,n))$ **alors**

$\text{Val}(f) \leftarrow \text{Val}(n) + k(f,n); \text{Pere}(f) \leftarrow n;$

si $(\text{Val}(f) < \text{Cout})$ **alors**

si (f est un nœud terminal) **alors**

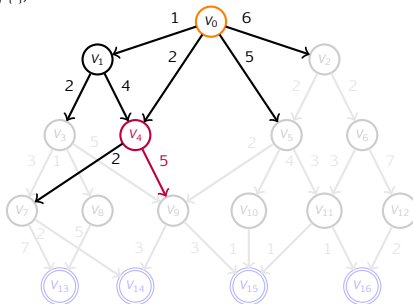
$\text{Cout} \leftarrow \text{Val}(f);$

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$n \text{ (Val}(n))$	$f \text{ (Val}(f))$	$\text{Pere}(f)$	V
$V_4(2)$	$V_7(4)$ $V_9(7)$	V_4 $\perp$	$\{V_3, V_7, V_5, V_2\}$ $\{V_3, V_7, V_5, V_2\}$

$\text{Cout} = +\infty, \text{Sol} = \{\}$



n (Val(n))	f (Val(f))	Pere(f)	V
V ₄ (2)	V ₇ (4) V ₉ (7)	V ₄ V₄	{V ₃ , V ₇ , V ₅ , V ₂ } {V ₃ , V ₇ , V ₅ , V ₂ }

$$\text{Cout} = +\infty, \text{Sol} = \{\}$$

1. **6**

2 5 $\rightarrow V_2$



Branch & Bound(Racine,Sol,Cout)

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si $(\text{Val}(f) > \text{Val}(n) + k(f,n))$ **alors**

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si $(\text{Val}(f) < \text{Cout})$ **alors**

si (f est un nœud terminal) **alors**

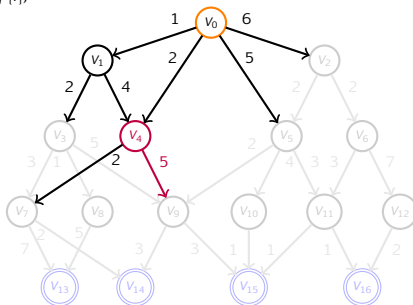
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$V_4(2)$	$V_7(4)$	V_4	$\{V_3, V_7, V_5, V_2\}$
	$V_9(7)$	V_4	$\{V_3, V_7, V_5, V_2\}$

$\text{Cout} = +\infty, \text{Sol} = \{\}$



n (Val(n))	f (Val(f))	Pere(f)	V
V ₄ (2)	V ₇ (4)	V ₄	{V ₃ , V ₇ , V ₅ , V ₂ }
	V ₉ (7)	V ₄	{V ₃ , V ₇ , V ₅ , V ₂ }

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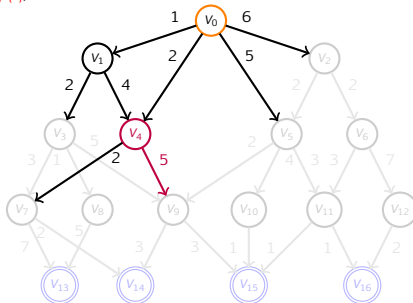
si (f est un nœud terminal) alors

Cout $\leftarrow$ Val(f);

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Sol ← cheminMenantA(f);
```

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sinon  $V \leftarrow V \cup_n \{f\};$ 
```

n (Val(n))	f (Val(f))	Pere(f)	V
V ₄ (2)	V ₇ (4) V ₉ (7)	V ₄ V ₄	{V ₃ , V ₇ , V ₅ , V ₂ } {V ₃ , V ₇ , V ₅ , V ₂ , V₉ }

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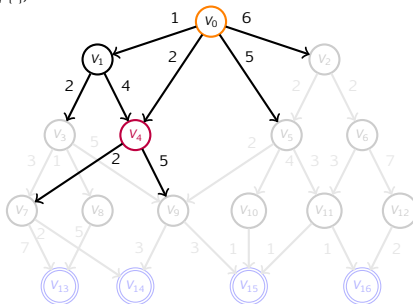
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$n \text{ (Val}(n))$	$f \text{ (Val}(f))$	$\text{Pere}(f)$	V
$V_4(2)$	$V_7(4)$ $V_9(7)$	V_4 V_4	$\{V_3, V_7, V_5, V_2\}$ $\{V_3, V_7, V_5, V_2, V_9\}$

$\text{Cout} = +\infty, \text{Sol} = \{\}$



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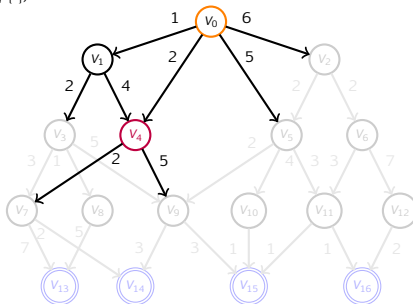
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n (Val(n))	f (Val(f))	Pere(f)	V
V ₄ (2)	V ₇ (4) V ₉ (7)	V ₄ V ₄	{V ₃ , V ₇ , V ₅ , V ₂ } {V ₃ , V ₇ , V ₅ , V ₂ , V ₉ }

$$\text{Cout} = +\infty, \text{Sol} = \{\}$$


n (Val(n))	f (Val(f))	Pere(f)	V
V ₃ (3)	()		{V ₇ , V ₅ , V ₂ , V ₉ }

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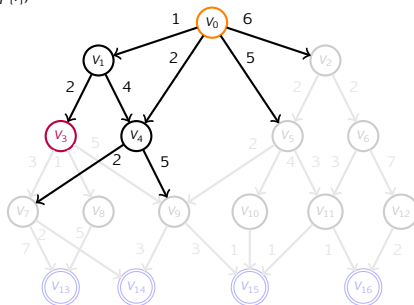
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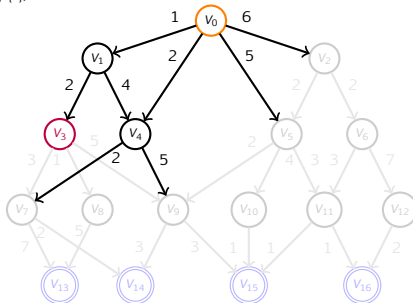
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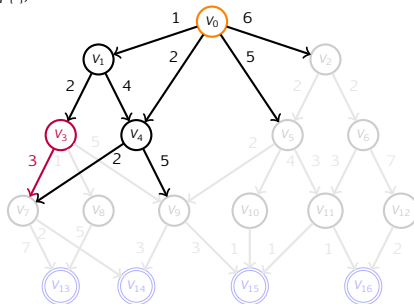
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V ₃ (3)	()		{V ₇ , V ₅ , V ₂ , V ₉ }

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V ₃ (3)	V ₇ (4)	V ₄	{V ₇ , V ₅ , V ₂ , V ₉ }

$$\text{Cout} = +\infty, \text{Sol} = \{\}$$
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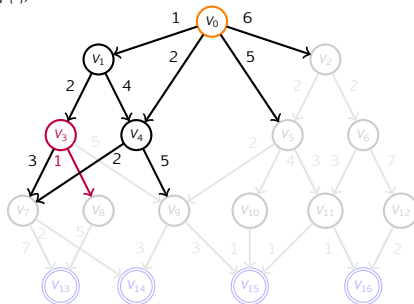
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n (Val(n))	f (Val(f))	Pere(f)	V
$V_3(3)$	$V_7(4)$ $V_8(0)$	V_4 $\perp$	$\{V_7, V_5, V_2, V_9\}$ $\{V_7, V_5, V_2, V_9\}$

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Branch & Bound(Racine,Sol,Cout)

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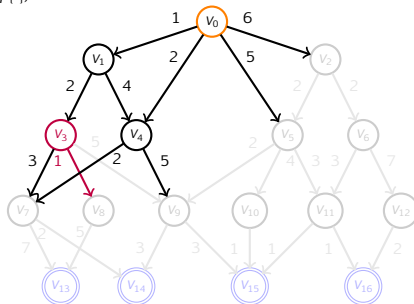
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n (Val(n))	f (Val(f))	Pere(f)	V
$V_3(3)$	$V_7(4)$ $V_8(0)$	V_4 $\perp V_3$	$\{V_7, V_5, V_2, V_9\}$ $\{V_7, V_5, V_2, V_9\}$

$$\text{Cout} = +\infty, \text{Sol} = \{\}$$


n (Val(n))	f (Val(f))	Pere(f)	V
V ₃ (3)	V ₇ (4) V ₈ (4)	V ₄ V ₃ V ₃	{V ₇ , V ₅ , V ₂ , V ₉ } {V ₇ , V ₅ , V ₂ , V ₉ }

$$\text{Cout} = +\infty, \text{Sol} = \{\}$$


2. $\rightarrow$ 5 $\rightarrow$ V_2



n (Val(n))	f (Val(f))	Pere(f)	V
V ₃ (3)	V ₇ (4)	V ₄	{V ₇ , V ₅ , V ₂ , V ₉ }
	V ₈ (4)	V ₃	{V ₇ , V ₅ , V ₂ , V ₉ }

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n (Val(n))	f (Val(f))	Pere(f)	V
V ₃ (3)	V ₇ (4)	V ₄	{V ₇ , V ₅ , V ₂ , V ₉ }
	V ₈ (4)	V ₃	{V ₇ , V ₅ , V ₂ , V ₉ }

$$\text{Cout} = +\infty, \text{Sol} = \{\}$$
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Branch & Bound(Racine,Sol,Cout)

$$\forall i \text{ val}(i) \leftarrow +\infty; \text{Pere}(i) \leftarrow \perp;$$
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tant que $V \neq \{\}$ faire

$$n \leftarrow \text{prendreUnElement}(V); V \leftarrow V \setminus \{n\};$$

si $(Val(n) < Cout)$ alors

pour chaque (fils f de n) faire

si $(Val(f) > Val(n) + k(f,n))$ alors

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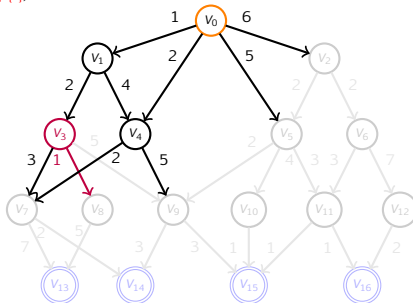
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Cout $\leftarrow$ Val(f);

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n (Val(n))	f (Val(f))	Pere(f)	V
V ₃ (3)	V ₇ (4) V ₈ (4)	V ₄ V ₃	{V ₇ , V ₅ , V ₂ , V ₉ } {V ₇ , V₈ , V ₅ , V ₂ , V ₉ }

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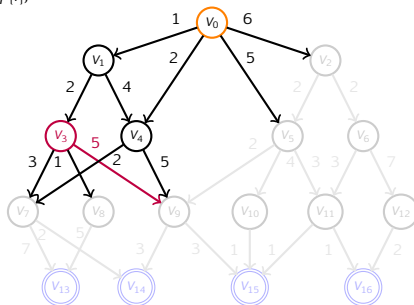
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	$V_8(4)$	V_3	$\{V_7, V_8, V_5, V_2, V_9\}$
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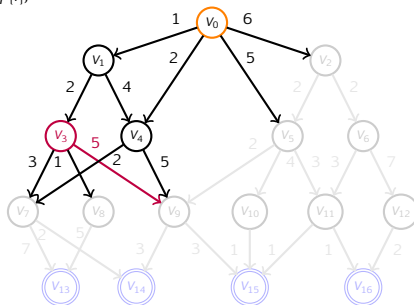
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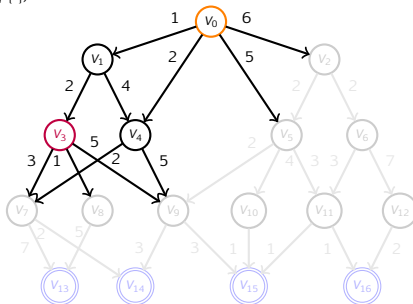
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V ₃ (3)	V ₇ (4)	V ₄	{V ₇ , V ₅ , V ₂ , V ₉ }
	V ₈ (4)	V ₃	{V ₇ , V ₈ , V ₅ , V ₂ , V ₉ }
	V ₉ (7)	V ₄	{V ₇ , V ₈ , V ₅ , V ₂ , V ₉ }

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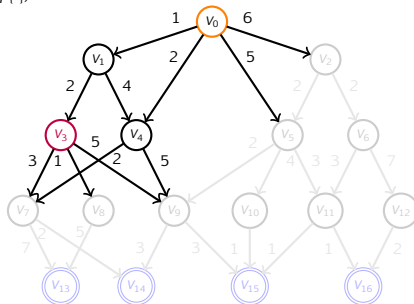
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	V ₉ (7)	V ₄	{V ₇ , V ₈ , V ₅ , V ₂ , V ₉ }

$$\text{Cout} = +\infty, \text{Sol} = \{\}$$


n (Val(n))	f (Val(f))	Pere(f)	V
V ₇ (4)	()		{V ₈ , V ₅ , V ₂ , V ₉ }

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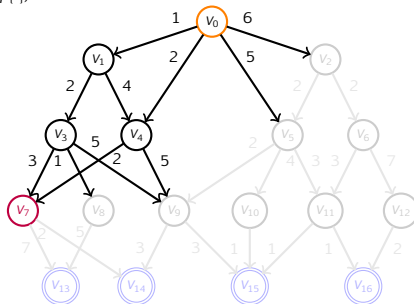
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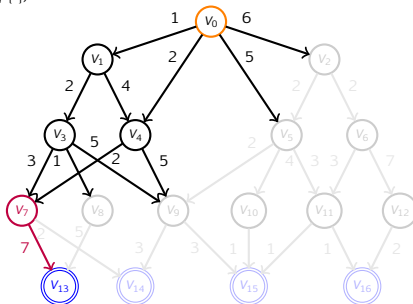
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n (Val(n))	f (Val(f))	Pere(f)	V
V ₇ (4)	V ₁₃ (0)	⊥	{V ₈ , V ₅ , V ₂ , V ₉ }

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n (Val(n))	f (Val(f))	Pere(f)	V
V ₇ (4)	V ₁₃ (11)	V ₇	{V ₈ , V ₅ , V ₂ , V ₉ }

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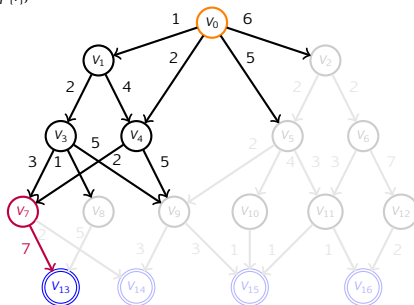
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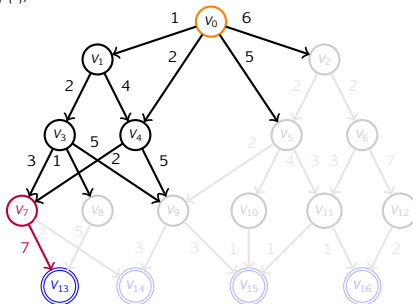
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Cout = 11, Sol = $\{V_0, V_4, V_7, V_{13}\}$



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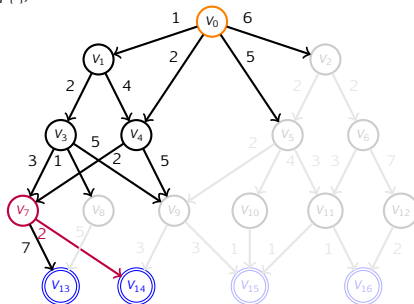
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n (Val(n))	f (Val(f))	Pere(f)	V
$V_7(4)$	$V_{13}(11)$ $V_{14}(0)$	V_7 $\perp$	$\{V_8, V_5, V_2, V_9\}$ $\{V_8, V_5, V_2, V_9\}$

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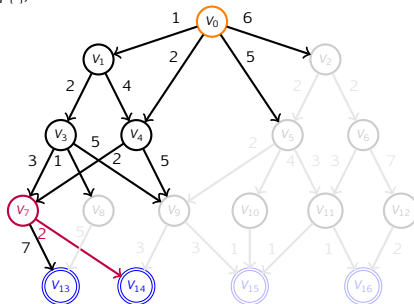
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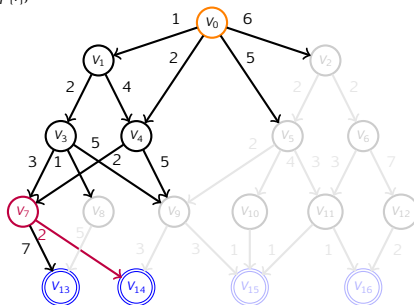
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$V_7(4)$	$V_{13}(11)$	V_7	$\{V_8, V_5, V_2, V_9\}$
	$V_{14}(6)$	V_7	$\{V_8, V_5, V_2, V_9\}$

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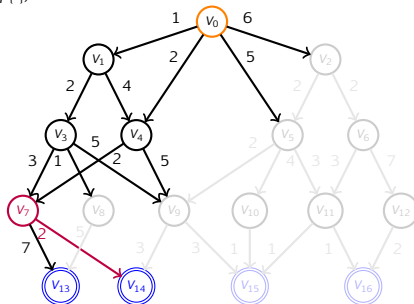
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si $(\text{Val}(n) < \text{Cout})$ **alors**

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si $(\text{Val}(f) > \text{Val}(n) + k(f,n))$ **alors**

$\text{Val}(f) \leftarrow \text{Val}(n) + k(f,n); \text{Pere}(f) \leftarrow n;$

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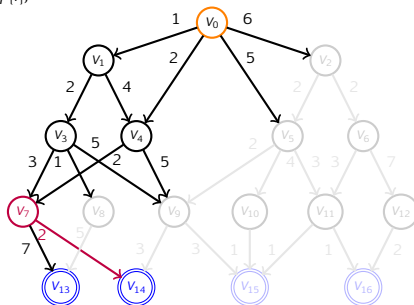
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$\text{Sol} \leftarrow \text{cheminMenantA}(f);$

sinon $V \leftarrow V \cup \{f\};$

n (Val(n))	f (Val(f))	Pere(f)	V
$V_7(4)$	$V_{13}(11)$	V_7	$\{V_8, V_5, V_2, V_9\}$
	$V_{14}(6)$	V_7	$\{V_8, V_5, V_2, V_9\}$

$\text{Cout} = 11, \text{Sol} = \{V_0, V_4, V_7, V_{13}\}$



Branch & Bound(Racine,Sol,Cout)

$\forall i \text{ val}(i) \leftarrow +\infty; \text{Pere}(i) \leftarrow \perp;$

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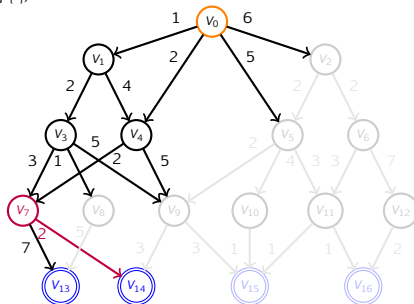
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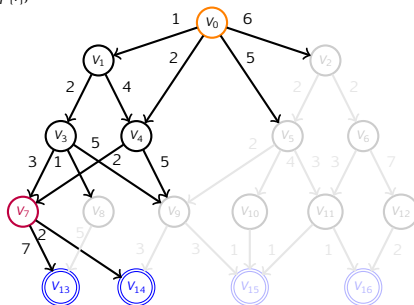
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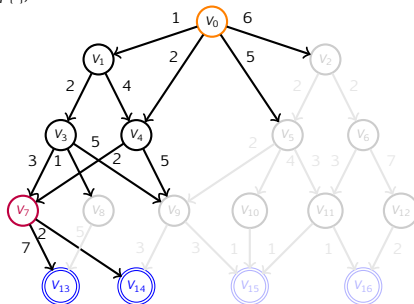
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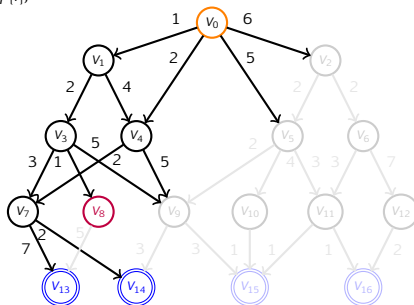
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$V_8(4)$	$()$		$\{V_5, V_2, V_9\}$

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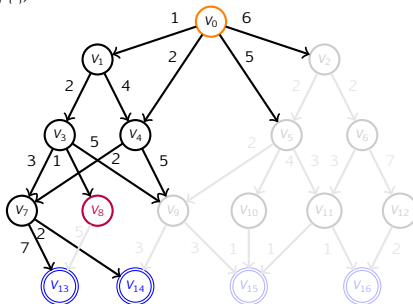
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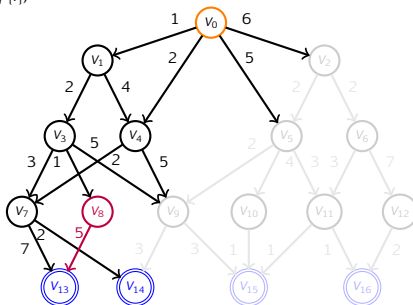
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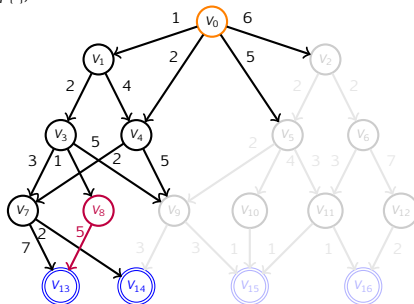
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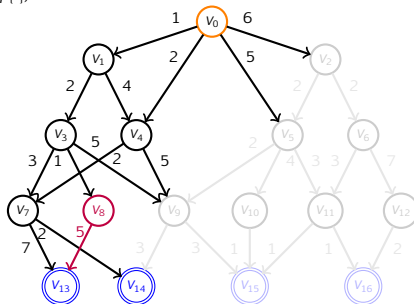
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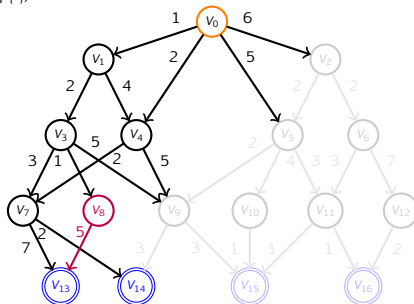
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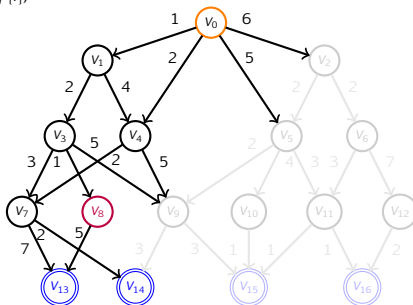
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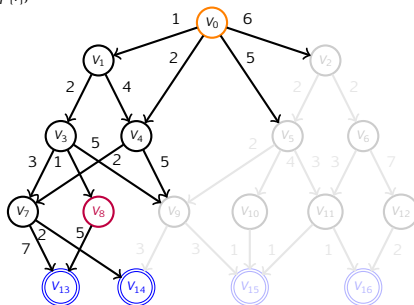
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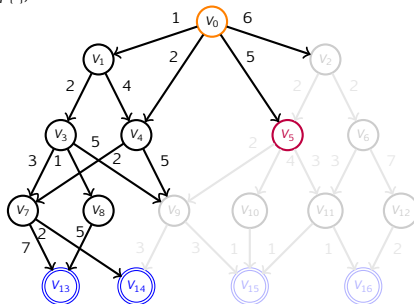
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n (Val(n))	f (Val(f))	Pere(f)	V
V ₅ (5)			{V ₂ , V ₉ }

$$\text{Cout} = 6, \text{Sol} = \{V_0, V_4, V_7, V_{14}\}$$
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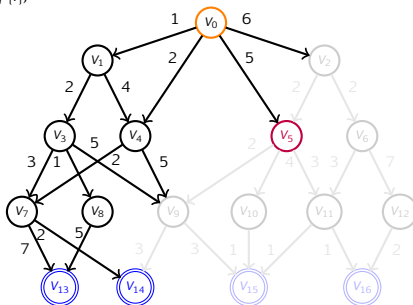
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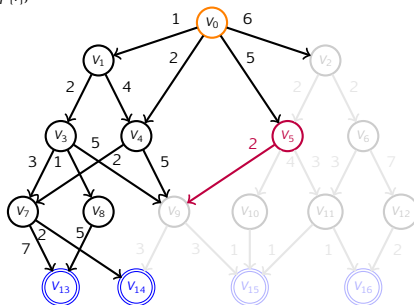
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n (Val(n))	f (Val(f))	Pere(f)	V
$V_5(5)$			$\{V_2, V_9\}$

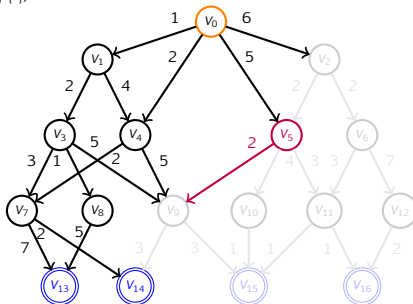
$\text{Cout} = 6, \text{Sol} = \{V_0, V_4, V_7, V_{14}\}$



n (Val(n))	f (Val(f))	Pere(f)	V
V ₅ (5)	V ₉ (7)	V ₄	{V ₂ , V ₉ }

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V ₅ (5)	V ₉ (7)	V ₄	{V ₂ , V ₉ }

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Branch & Bound(Racine,Sol,Cout)

$\forall i \text{ val}(i) \leftarrow +\infty; \text{Pere}(i) \leftarrow \perp;$

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tant que $V \neq \{\}$ **faire**

$n \leftarrow \text{prendreUnElement}(V); V \leftarrow V \setminus \{n\};$

si $(\text{Val}(n) < \text{Cout})$ **alors**

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si $(\text{Val}(f) > \text{Val}(n) + k(f,n))$ **alors**

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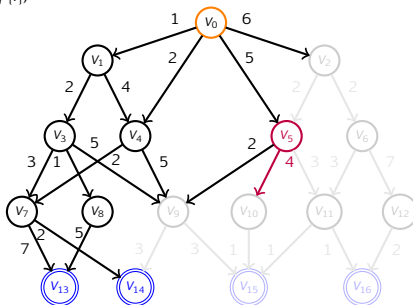
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$n \text{ (Val}(n))$	$f \text{ (Val}(f))$	$\text{Pere}(f)$	V
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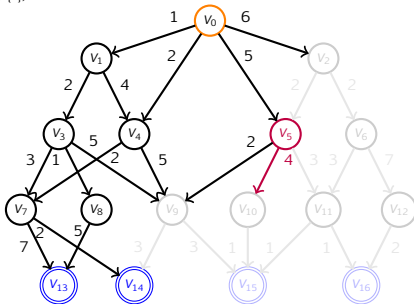
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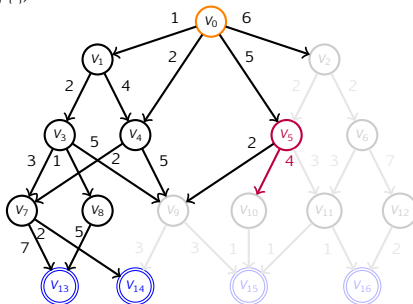
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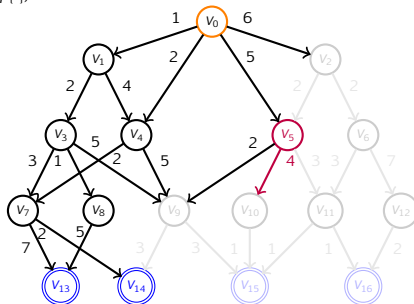
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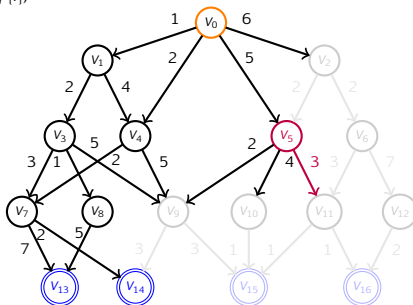
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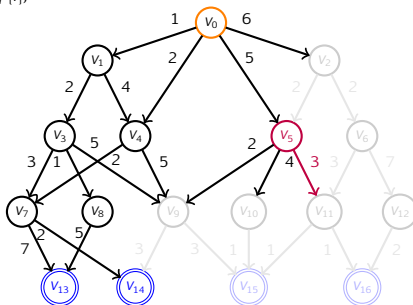
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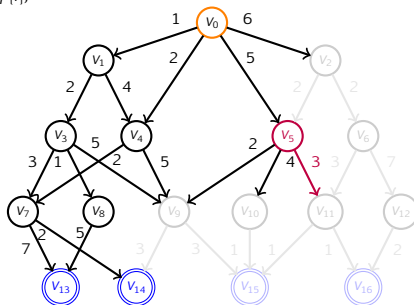
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n (Val(n))	f (Val(f))	Pere(f)	V
V ₅ (5)	V ₉ (7)	V ₄	{V ₂ , V ₉ }
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	V ₁₁ (8)	V ₅	{V ₂ , V ₉ }

$$\text{sinon } V \leftarrow V \cup_n \{f\};$$

The graph shows a binary tree structure with 17 nodes. The root node is V_0 (orange). It has two children: V_1 (left) and V_2 (right). V_1 has children V_3 and V_4 . V_2 has children V_5 and V_6 . V_3 has children V_7 and V_8 . V_4 has children V_9 and V_{10} . V_5 has children V_{11} and V_{12} . V_6 has children V_{13} and V_{14} . V_7 has children V_{15} and V_{16} . V_8 has children V_{17} and V_{18} . The edges are labeled with weights. A red arrow points from V_5 to V_{11} with weight 3.

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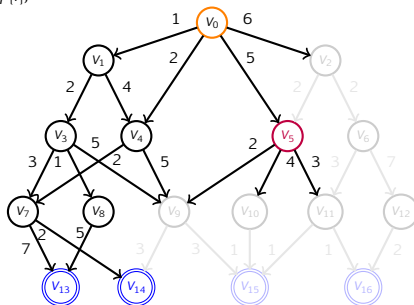
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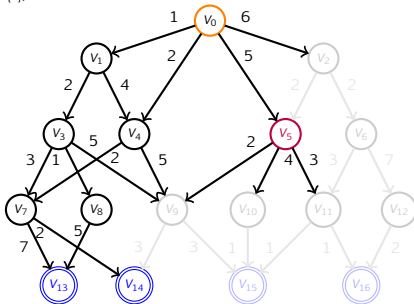
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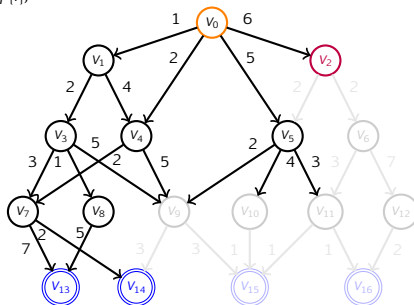
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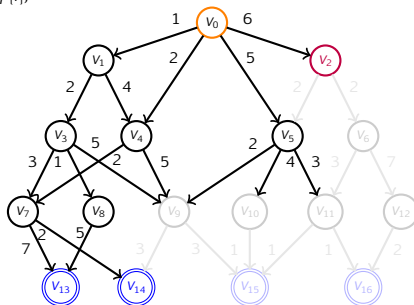
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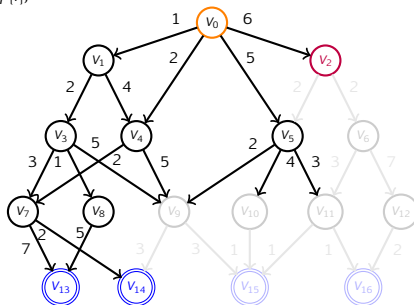
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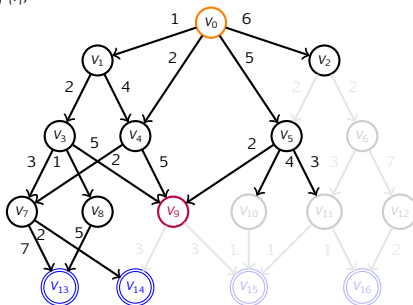
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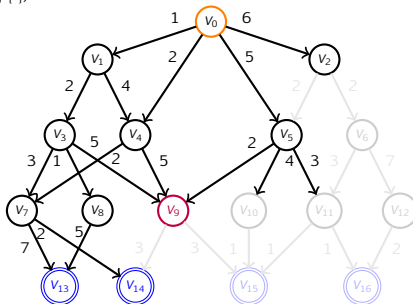
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$V_9(7)$			$\{\}$

$\text{Cout} = 6, \text{Sol} = \{V_0, V_4, V_7, V_{14}\}$



Branch & Bound(Racine,Sol,Cout)

$\forall i \text{ val}(i) \leftarrow +\infty; \text{Pere}(i) \leftarrow \perp;$

$\text{Cout} \leftarrow +\infty; \text{Sol} \leftarrow \{\};$

$V \leftarrow \{\text{Racine}\}; \text{Val}(\text{Racine}) \leftarrow 0;$

tant que $V \neq \{\}$ **faire**

$n \leftarrow \text{prendreUnElement}(V); V \leftarrow V \setminus \{n\};$

si $(\text{Val}(n) < \text{Cout})$ **alors**

pour chaque (fils f de n) **faire**

si $(\text{Val}(f) > \text{Val}(n) + k(f,n))$ **alors**

$\text{Val}(f) \leftarrow \text{Val}(n) + k(f,n); \text{Pere}(f) \leftarrow n;$

si $(\text{Val}(f) < \text{Cout})$ **alors**

si (f est un nœud terminal) **alors**

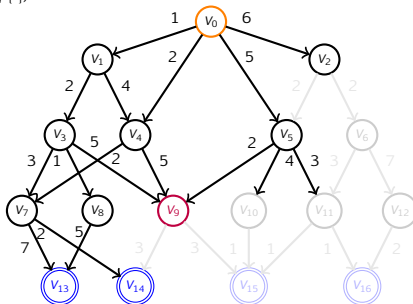
$\text{Cout} \leftarrow \text{Val}(f);$

$\text{Sol} \leftarrow \text{cheminMenantA}(f);$

sinon $V \leftarrow V \cup \{f\};$

n (Val(n))	f (Val(f))	Pere(f)	V
$V_9(7)$			$\{\}$

$\text{Cout} = 6, \text{Sol} = \{V_0, V_4, V_7, V_{14}\}$



Branch & Bound(Racine,Sol,Cout)

n (Val(n))	f (Val(f))	Pere(f)	V

$\forall i \text{ val}(i) \leftarrow +\infty$; $\text{Pere}(i) \leftarrow \perp$;

$\text{Cout} \leftarrow +\infty$; $\text{Sol} \leftarrow \{\}$;

$V \leftarrow \{\text{Racine}\}$; $\text{Val}(\text{Racine}) \leftarrow 0$;

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$n \leftarrow \text{prendreUnElement}(V)$; $V \leftarrow V \setminus \{n\}$;

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si ($\text{Val}(f) < \text{Cout}$) **alors**

si (f est un nœud terminal) **alors**

$\text{Cout} \leftarrow \text{Val}(f)$;

$\text{Sol} \leftarrow \text{cheminMenantA}(f)$;

sinon $V \leftarrow V \cup \{f\}$;

Solution $\{V_0, V_4, V_7, V_{14}\}$ avec un coût de 6

